Validation of a finite-element solver for one-dimensional viscous sheet retraction

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Abstract

We validate a finite-element solver for the one-dimensional (long-wave) retraction of a viscous liquid sheet against an independent finite-difference implementation and against the asymptotic results of Ahsan *et al.* (arXiv:2507.04603). The two solvers share no code, no discretisation and no coordinate system, and agree to between 2×10^{-5} and 1.3×10^{-4} in the retraction speed across four decades of time. Measured against the published asymptotics, the early-time law and the late-time finite-sheet law are reproduced without adjustable constants, as are the tip-thickness relation and the crossover time $T_c = \mathcal{L}$. In the late-time infinite-sheet regime we find that the speed deficit is well described by $1 - \mathrm{d}X_c/\mathrm{d}T \simeq \frac{3}{2}(1+T)^{-2}$, where the published coefficient is unity; the coefficient is stated in that work to be a numerical fit rather than a result of the asymptotic analysis.

1 Scope and nomenclature

Two solvers are compared throughout.

TC1D-FEM is the solver under validation: a finite-element discretisation of the long-wave equations in a Lagrangian mass coordinate, with the sheet thickness h on a cell-constant space and the velocity u on continuous piecewise-linear elements, integrated with BDF2. The free edge is a material point of the mesh, so no interface tracking or remeshing is required.

ABDS-CN is an independent reference: a transcription into Python of the finite-difference solver released with Ahsan *et al.*, which advances a Crank–Nicolson discretisation on a static grid with Picard iteration and a tridiagonal solve. Appendix A establishes that the transcription reproduces the original code to the precision at which the original can be read.

The asymptotic predictions of Ahsan *et al.* are treated here as hypotheses to be tested, and are plotted as such: they are not used to normalise or calibrate either solver.

1.1 Units

Ahsan *et al.* scale on the half-thickness $h_0 = H/2$, with $u_{TC} = \sqrt{\gamma/\rho h_0}$, Ohnesorge number $Oh_p = \mu/\sqrt{\rho h_0 \gamma}$ and visco-inertial time $t_I = 4\mu h_0/\gamma$. This work scales on the full thickness H, with the Savva–Bush Ohnesorge number Oh and viscous time $\tau_{\mathrm{vis}} = \mu H/2\gamma$. The maps are

$$Oh_p = 2\,Oh, \qquad T = t^*/4, \qquad \mathcal{L} = \frac{\ell_0}{4h_0 Oh_p} = \frac{M}{4\,Oh}, \tag{1}$$

where $t^* = t/\tau_{\mathrm{vis}}$ and M is the total sheet mass per unit width, which is the extent of the computational domain in the mass coordinate. Their retraction speed $\mathrm{d}X_c/\mathrm{d}T$ equals $u^* = U/V_{TC}$ with no intervening factor. These relations were verified symbolically, and independently confirmed by two physical identities: their tip condition $HV_X = -1$ is the natural boundary condition of the present formulation, and their $H(X_c, T) = 1 + T$ is the tip-thickness relation measured here as $h_{\mathrm{tip}} = 1 + t^*/4$.

Equation (1) also shows that the present formulation represents a sheet of finite length: $\mathcal{L}$ is set by the mass domain M, and the far boundary condition (vanishing velocity, free thickness) is the condition imposed in the reference work. The mapping was checked before use: at $M = 20$, $Oh = 1$, the deceleration is observed at $T_c = 4.97$ against $\mathcal{L} = 5$.

2 Early time: $T \ll 1$

At early times the reference work predicts a diffusive growth of the retraction speed, $\mathrm{d}X_c/\mathrm{d}T \sim 2\sqrt{T/\pi}$ (their eq. 5.10). Figure 1 compares both solvers and that prediction at $\mathcal{L} = 0.1$, 1 and 20.

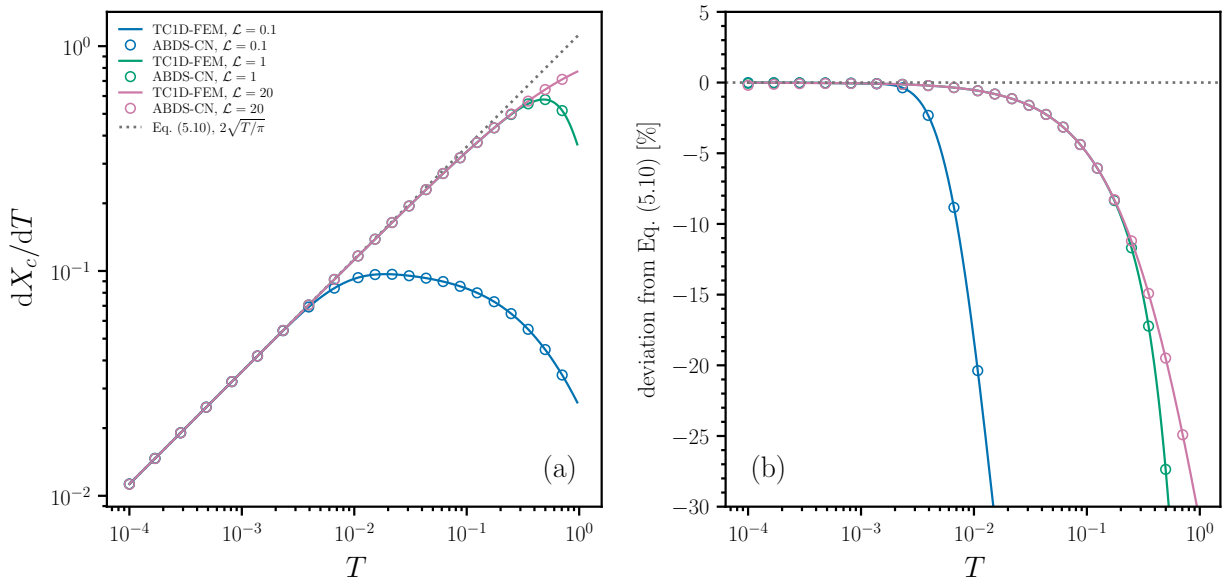


Figure 1: Retraction speed at early time. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: the diffusive prediction $2\sqrt{T/\pi}$.

The prediction is reproduced to better than 0.06% for $T \leq 10^{-3}$ at every $\mathcal{L}$. Its range of validity depends strongly on $\mathcal{L}$: taking the time at which the relative deviation first exceeds 1%, 5% and 10%,

	1%	5%	10%
$\mathcal{L} = 0.1$	$T = 0.0031$	0.0053	0.0075
$\mathcal{L} = 1$	$T = 0.019$	0.102	0.213

The two solvers agree over this window to a maximum absolute difference of 2.4×10^{-5} at $\mathcal{L} = 0.1$ and 1.3×10^{-4} at $\mathcal{L} = 1$.

3 Late time at finite length: the deceleration

When the free edge approaches the far end of the sheet, $\mathcal{L} - X_c \ll 1$, the reference work predicts $\mathrm{d}X_c/\mathrm{d}T \sim \mathcal{L}(1+T)^{-2}$ (their eq. 6.10). This regime is the principal focus of that work.

The prediction is reproduced without any adjustable constant. At the end of each run the ratio of the computed speed to eq. (6.10) lies between 1.00002 and 1.0010, and the remaining sheet length $\mathcal{L} - X_c$ agrees with $\mathcal{L}/(1+T)$ to 1.3×10^{-4}. Continuing the $\mathcal{L} = 5$ case to $T = 400$ shows the relative departure decaying as $O(T^{-2})$, which is the order of the neglected terms.

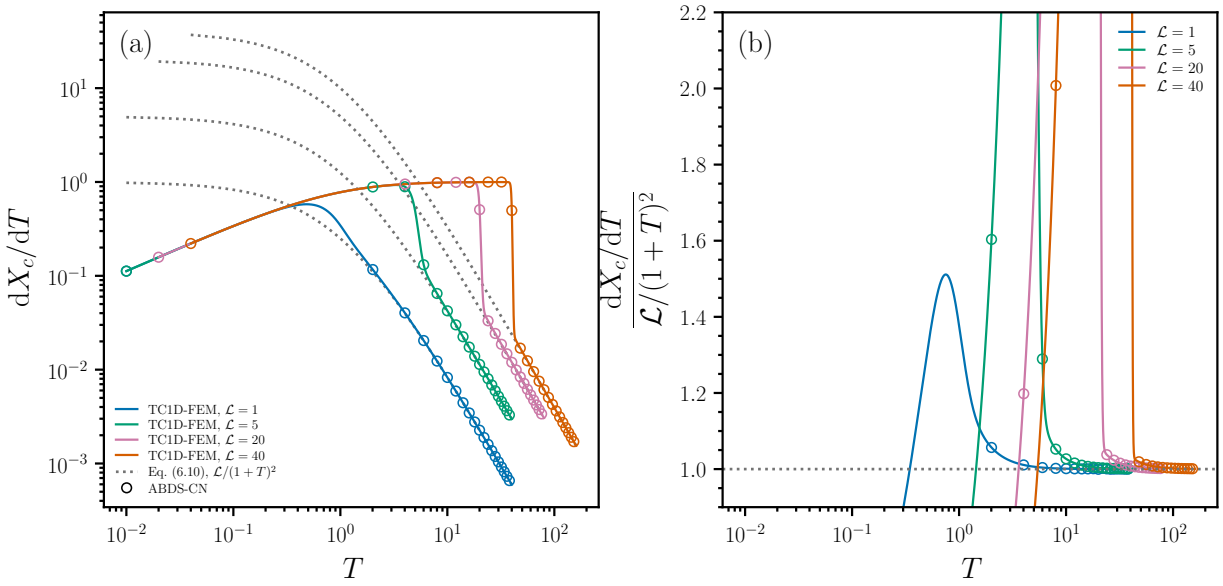


Figure 2: Deceleration at finite $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: $\mathcal{L}(1+T)^{-2}$.

4 Late time at large length

The regime of principal interest here is $\mathcal{L} \gg T \gg 1$, in which the edge has reached the Taylor–Culick speed but has not yet sensed the far end. The reference work gives, for an infinite sheet,

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} \sim 1 - \frac{1}{(1+T)^2}. \tag{2}$$

Two points are worth stating before the comparison. The deficit is $(1+T)^{-2}$, not a pure power T^{-2}; fitting a power law in T and comparing exponents is therefore not a test of eq. (2). And the coefficient is described in that work as a numerical fit, with the corresponding expansion coefficient stated to remain undetermined by the asymptotic analysis and to depend on the history of the retraction.

4.1 The measured deficit

Figure 3 shows the speed and the deficit $1 - \mathrm{d}X_c/\mathrm{d}T$. Forming the ratio $(1+T)^2\big(1 - \mathrm{d}X_c/\mathrm{d}T\big)$, which eq. (2) requires to approach unity, both solvers give

T	10	20	30	40	50
ABDS-CN (Richardson $\Delta X \to 0$)	1.4391	1.4891	1.4959	1.4978	1.4985
TC1D-FEM ($\mathcal{L} = 200$)	1.4382	1.4889	1.4959	1.4977	1.4986
TC1D-FEM ($\delta t/2$)	1.4382	1.4889	1.4959	1.4977	1.4986
TC1D-FEM (mesh $\times 2$)	1.4383	1.4889	1.4960	1.4978	1.4986

The ratio approaches 3/2. Extrapolating the plateau with several fit forms gives 1.5006, 1.5053 and 1.4977, from which we take

$$-U_{-2} = 1.500 \pm 0.003, \tag{3}$$

consistent with exactly 3/2 and inconsistent with unity.

The result is insensitive to every numerical parameter varied: halving the timestep changes it by 1.3×10^{-8} and doubling the mesh by 1.6×10^{-6}, against a deficit of 5.8×10^{-4} at $T = 50$.

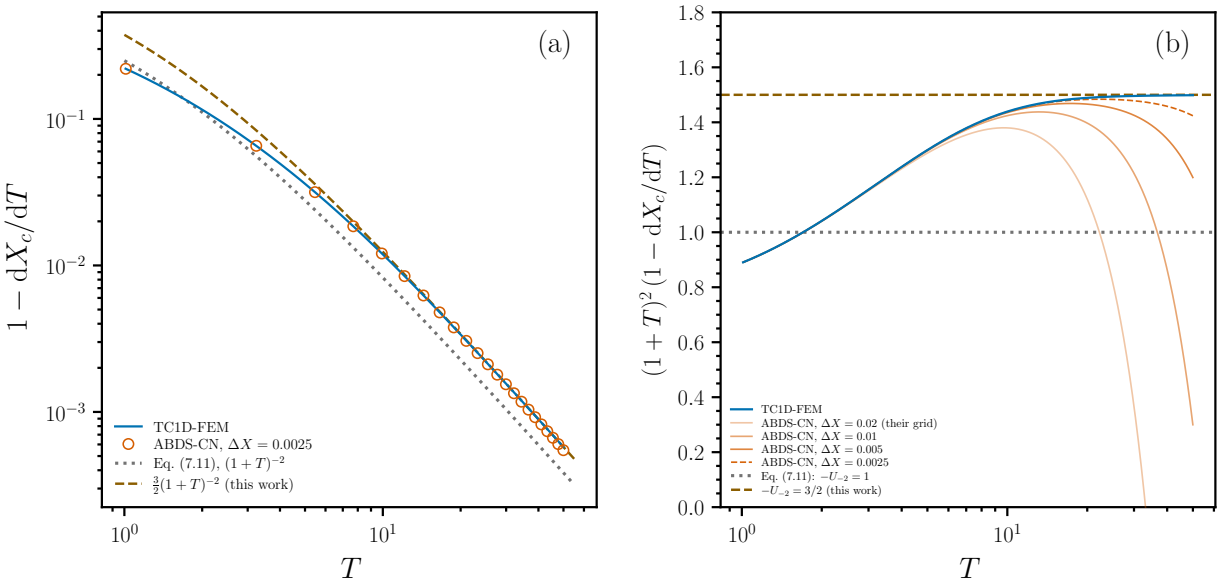


Figure 3: Late-time speed and deficit at large $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: eq. (2).

ABDS-CN converges at an observed order of 2.000–2.002 in ΔX, and its Richardson limit agrees with TC1D-FEM to 4.8×10^{-5}.

A second, independent check supports the same conclusion. The reference work's eq. (7.17) gives $X_c \sim T + 1$, whereas its own eqs. (7.8) and (7.10), its eqs. (7.13) and (7.16), and direct integration of eq. (2) all give $X_c \sim T - 1$. Both solvers measure $X_c(50) - 50 = -0.971$. The relation $X_c \sim T - 1$ is equivalent to the exact statement $\int_0^\infty \left(1 - \mathrm{d}X_c/\mathrm{d}T\right) \mathrm{d}T = 1$.

4.2 A two-term form

A form consistent with the measured coefficient and with the integral constraint is

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} \simeq 1 - \frac{3}{2}\frac{1}{(1+T)^2} + \frac{1}{(1+T)^3}, \tag{4}$$

which reproduces the numerics to within 2.7% for $T \geq 5$, against 33% for eq. (2) over the same range. We emphasise that eq. (4) is constructed to match a measured coefficient and an exact constraint; it is not derived.

4.3 Profiles and the tip thickness

Figure 4 compares thickness and velocity profiles against the reference work's eqs. (7.13) and (7.14). The tip-thickness relation $H(X_c, T) = 1 + T$ is reproduced to a maximum relative deviation of 3.6×10^{-5} over $1 \leq T \leq 50$. The profile predictions agree to 0.043% and 1.55% respectively at $T = 50$, improving monotonically with T.

5 The crossover time

For large $\mathcal{L}$ the edge decelerates from the Taylor–Culick speed onto the finite-sheet law of §3. The reference work reports the crossover time T_c and its duration ΔT_c as functions of $\mathcal{L}$, with $T_c = \mathcal{L}$ for large $\mathcal{L}$.

Neither quantity is defined in closed form in that work, so an operational definition is adopted here and applied identically to both solvers; it is stated with the figure. Figure 5 compares both solvers against values digitised from the published figure.

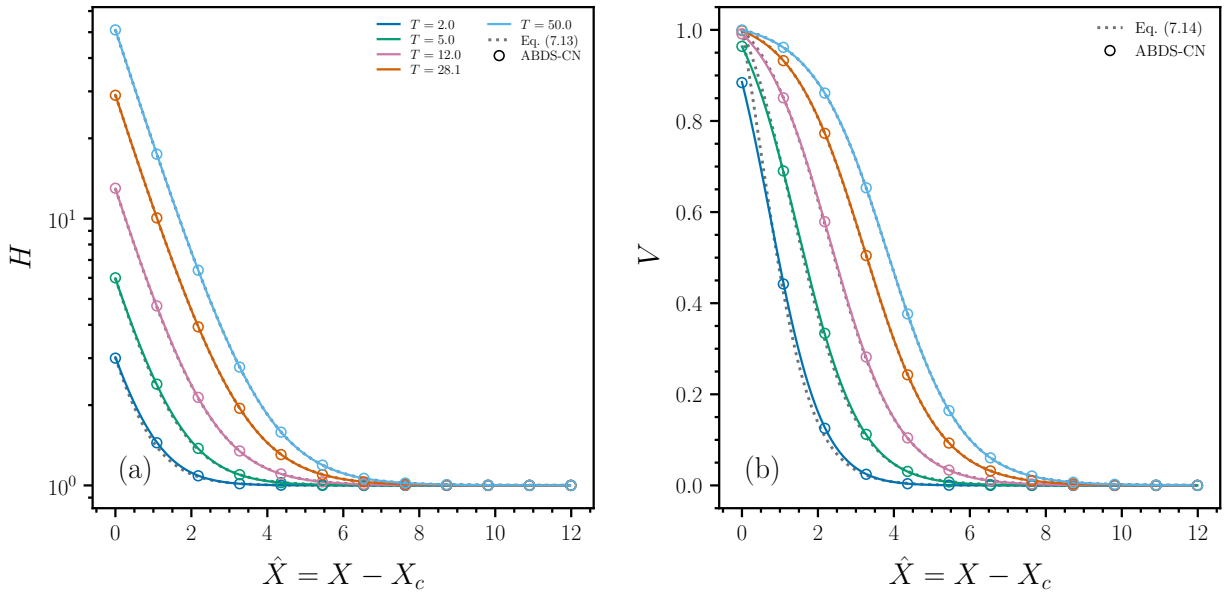


Figure 4: Thickness and velocity profiles at large $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: eqs. (7.13) and (7.14).

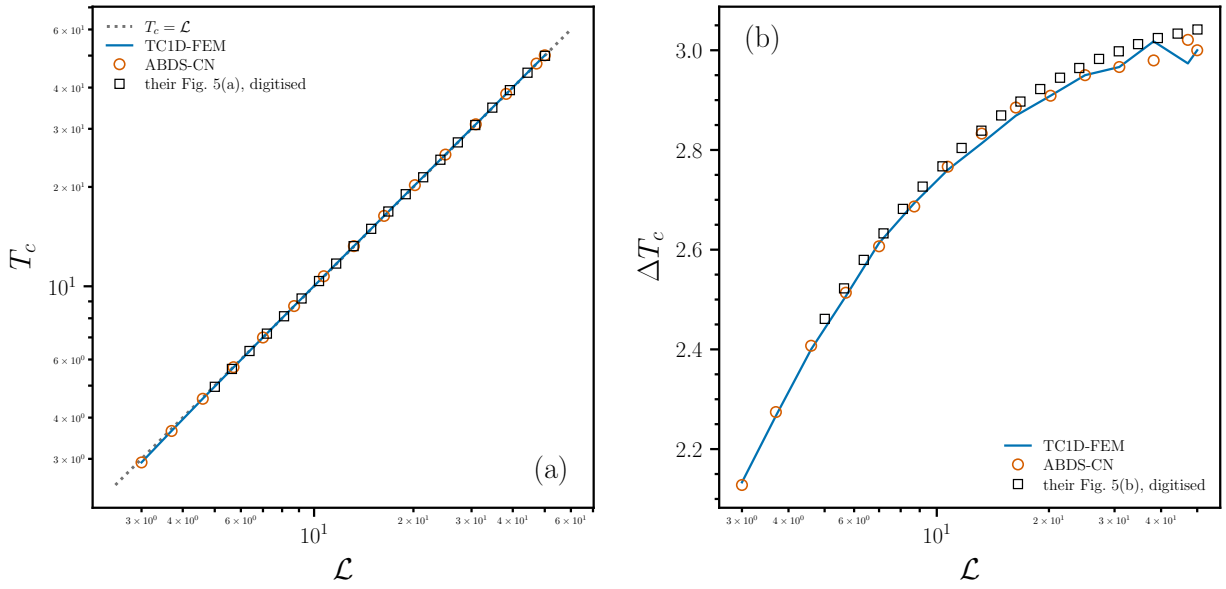


Figure 5: Crossover time T_c and duration ΔT_c against $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Circles: values digitised from the published figure.

We find $T_c/\mathcal{L} = 1$ to within 0.25% for $\mathcal{L} \geq 7$. Against the digitised values, T_c agrees to 0.142% RMS, within the estimated digitisation uncertainty of 0.2%, and ΔT_c to 0.024 absolute (1.5%).

One point of definition should be recorded. The text of the reference work describes ΔT_c as a full width at half maximum, while the released code computes twice that width. The published values are consistent with the code. Both conventions are reported here.

A Fidelity of the reference transcription

ABDS-CN is a transcription of released MATLAB into Python. To establish that the transcription is faithful rather than merely plausible, the original code was executed under GNU Octave and compared with the transcription at identical parameters. Over six configurations the largest absolute difference in the retraction speed is 5.0×10^{-10}, and the largest normalised difference across all twenty-four recorded quantities is 3.0×10^{-8}. The former is the resolution at which the original code writes its output, so the two agree to every digit that can be compared.

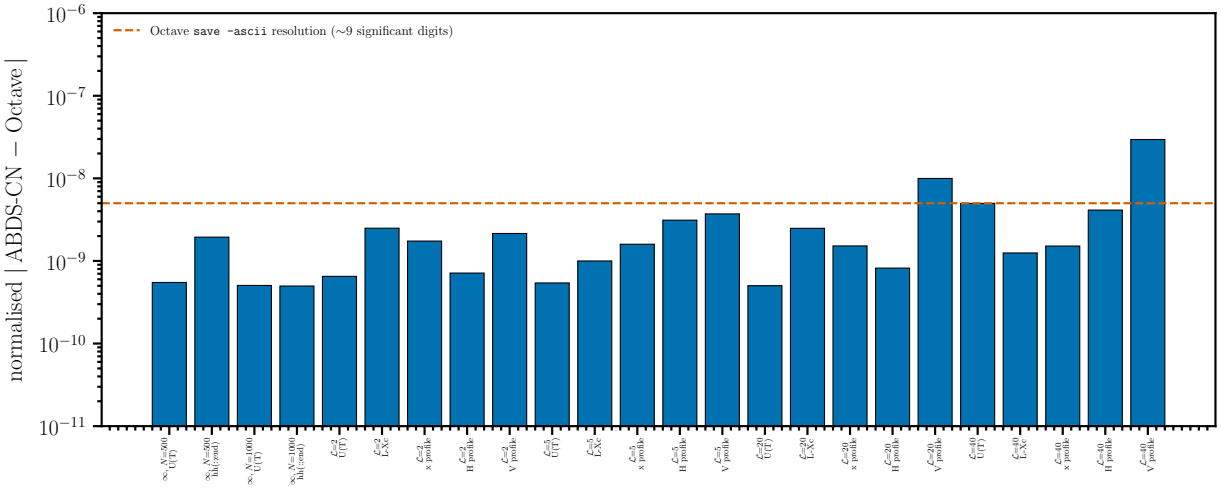


Figure 6: Original released code executed under GNU Octave, against the Python transcription ABDS-CN, at several $\mathcal{L}$.

B Reproduction

The campaign comprises 72 cases. Each is recorded in a manifest with its parameters and the exact command that regenerates it, alongside the derived series used for every figure and table in this document.